
Crash Course in Mathematics for Cognitive Science

Lecture Slides — Five Sessions

Probability & Statistics · Linear Algebra

Master of Cognitive Science — ENS Paris

Session 1 — Probability I

What today is about

Almost everything we study in cognitive science is *uncertain*: the next response of a participant, the number of spikes a neuron fires, whether a model gets an answer right.

Probability is simply the careful mathematics of that uncertainty. Today we build it from the ground up.

- Describe what can happen: *sample spaces* and *events*.
- Combine chances with a few simple *rules*.
- Update a belief when new evidence arrives: **Bayes' theorem**.
- Meet the handful of *distributions* that describe most real data.

The one idea to hold onto

Everything today leads to one destination.

If you remember only one thing

Bayes' theorem tells us how observing evidence should change what we believe. It is the mathematical backbone of Bayesian models of perception, learning, and decision-making — and of machine learning.

We will spend the first hour building the tools we need to understand it, and the second hour using it and meeting the common distributions.

A probability is just a number between 0 and 1

Before any formulas, fix the picture in your head.

- A probability measures our *degree of certainty* that something happens.
- 0 means impossible, 1 means certain, 0.5 means “as likely as not”.
- Everything else is rules for combining these numbers without contradicting ourselves.

Intuition

Think of one unit of “belief” that we spread across all the things that could happen. Probability is just how much of that belief sits on each outcome.

Why this matters for cognitive science

This is not only mathematics — it is a theory of the mind.

A leading view in modern cognitive science is that the brain is, in part, a *probabilistic inference machine*. Perception is treated as inferring the most probable cause of ambiguous sensory data; learning as updating beliefs from experience.

Link to cognitive science

Bayesian models of cognition, ideal-observer analysis, and the probabilistic models underlying machine learning all speak the language we build today. Learning it lets you read a large part of the modern literature.

Step 1: list everything that can happen

To reason about chance we first say, precisely, what the possible outcomes are.

Definition

The **sample space** Ω is the set of all possible outcomes of an experiment. An **event** A is a subset of Ω — a collection of outcomes we care about.

A few concrete examples:

- One die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$; the event “even” is $\{2, 4, 6\}$.
- A detection task: $\Omega = \{\text{yes}, \text{no}\}$.
- Spikes in one second: $\Omega = \{0, 1, 2, 3, \dots\}$.

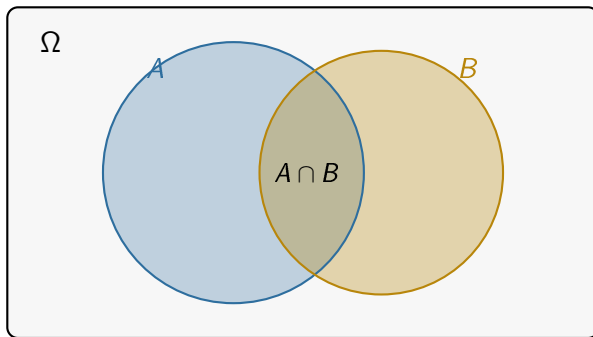
Events are yes/no questions about the outcome

An event is just a question with a yes/no answer: “was the die even?” splits Ω into the outcomes where the answer is yes and those where it is no.

Because events are sets, we combine them with the same words we use in ordinary logic:

- $A \cup B$ — “*A or B*”: outcomes in either (or both).
- $A \cap B$ — “*A and B*”: outcomes in both at once.
- A^c — “*not A*”: every outcome outside A .
- A, B are *disjoint* if they cannot both happen: $A \cap B = \emptyset$.

Picturing events: the Venn diagram



The whole rectangle is Ω . The overlap is “ A and B ”; the two circles together are “ A or B ”; everything outside both circles is “neither”.

When outcomes are equally likely, we just count

If every outcome is equally likely, probability becomes a counting problem:

$$\mathbb{P}(A) = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } \Omega}.$$

So the skill we need is counting outcomes efficiently. Two ideas cover almost everything.

Permutations vs. combinations

Order matters (permutations): $\frac{n!}{(n-k)!}$ ways to arrange k of n .

Order does not (combinations): $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ ways to choose k of n .

A reading group and a podium

A lab has 8 members.

How many 3-person reading groups? A group is a *set* — order does not matter — so we use a combination:

$$\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

How many gold–silver–bronze podiums? Now order *does* matter, so we use a permutation: $8 \cdot 7 \cdot 6 = 336$.

The podium counts each trio in all $3! = 6$ possible orders — which is exactly why the group count is $336/6 = 56$.

Practice — counting

Try these on your own; we will solve them together on the board.

Practice — try it

1. A questionnaire has 5 yes/no items. How many different response patterns are possible?
2. From a group of 10 participants, how many ways are there to form a 4-person focus group?
3. In how many different orders can 6 pictures be shown to a subject?

Three common-sense rules (the axioms)

Any sensible way of assigning probabilities must obey three rules. They simply formalise common sense.

Kolmogorov's axioms

- (1) $\mathbb{P}(A) \geq 0$ — no negative chances.
- (2) $\mathbb{P}(\Omega) = 1$ — something must happen.
- (3) If A and B are disjoint, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Everything else in probability can be derived from just these three statements.

Two rules you will use constantly

From the axioms come the two everyday tools.

Useful fact

Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Addition: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Why the subtraction? If we simply add $\mathbb{P}(A)$ and $\mathbb{P}(B)$, we count the overlap $A \cap B$ *twice* — once in each. Subtracting it once puts things right. If the events are disjoint there is no overlap, and the rule reduces to plain addition.

Addition rule — worked example

Example

In a study, 60% of participants are bilingual (B), 30% are musicians (M), and 20% are both.

Reading the answers straight off the rules:

- *Bilingual or musician*: $\mathbb{P}(B \cup M) = 0.60 + 0.30 - 0.20 = 0.70$.
- *Neither*: by the complement rule, $1 - 0.70 = 0.30$.
- *Bilingual but not a musician*: $\mathbb{P}(B) - \mathbb{P}(B \cap M) = 0.60 - 0.20 = 0.40$.

A trick worth its own slide: “at least one”

Questions about “at least one” are almost always easier through the *complement*: it is usually simpler to compute the chance of *none*.

Example

A target appears on each of 3 independent trials with probability 0.2.

The chance of *no* target on a single trial is 0.8, so across three independent trials

$$\mathbb{P}(\text{no target at all}) = 0.8^3 = 0.512,$$

and therefore $\mathbb{P}(\text{at least one target}) = 1 - 0.512 = 0.488$. Much easier than adding up the “exactly one / exactly two / exactly three” cases.

A pitfall to flag early

Common pitfall

The addition rule collapses to $\mathbb{P}(A) + \mathbb{P}(B)$ *only when the events are disjoint*. Forgetting the $-\mathbb{P}(A \cap B)$ term is the single most common early mistake.

A useful self-check: if a probability calculation ever gives you a number greater than 1, you have almost certainly double-counted an overlap somewhere. A probability can never exceed 1.

Practice — try it

In a sample, $\mathbb{P}(\text{coffee}) = 0.5$, $\mathbb{P}(\text{tea}) = 0.3$, and $\mathbb{P}(\text{both}) = 0.1$. Find:

1. $\mathbb{P}(\text{coffee or tea})$;
2. $\mathbb{P}(\text{neither})$;
3. $\mathbb{P}(\text{exactly one of the two})$.

Learning from information: conditional probability

The central question of the whole session: *how does knowing one thing change the probability of another?*

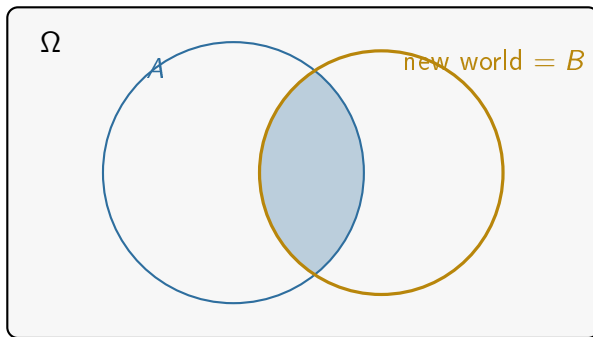
Definition

The **conditional probability** of A given B (with $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Read it as: “among the cases where B happened, how often does A also happen?”

Conditioning means zooming in



Conditioning on B throws away everything outside the B circle. Our new “world” is B , and $\mathbb{P}(A \mid B)$ is the fraction of that world covered by A . Dividing by $\mathbb{P}(B)$ rescales it so the shrunken world again totals 1.

The multiplication rule, and a quick example

Rearranging the definition gives a way to build the probability of “ A and B ” step by step:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B).$$

Example

Roll a fair die. What is $\mathbb{P}(\text{even} \mid \text{greater than } 3)$?

The condition “greater than 3” leaves the outcomes $\{4, 5, 6\}$. Among those, the even ones are $\{4, 6\}$. So

$$\mathbb{P}(\text{even} \mid > 3) = \frac{2}{3},$$

up from $\frac{1}{2}$ without the information. Knowing more changed the probability.

Averaging over scenarios: total probability

Often we know how likely A is *within* each of several scenarios, and want its overall probability.

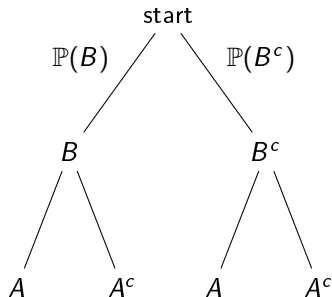
Law of total probability

If $B_1, \dots, B_k$ are disjoint and cover everything, then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

In words: work out how likely A is in each scenario, then take a weighted average, weighting each scenario by how likely it is. A picture makes this automatic — see the next slide.

The probability tree



Multiply probabilities along a branch, then *add* across the branches that end in A :

$$\mathbb{P}(A) = \mathbb{P}(B)\mathbb{P}(A \mid B) + \mathbb{P}(B^c)\mathbb{P}(A \mid B^c).$$

Total probability — worked example

Two boxes

Box 1 holds 3 red and 1 blue token; Box 2 holds 1 red and 3 blue. You pick a box at random (each with probability $\frac{1}{2}$) and draw one token.

What is $\mathbb{P}(\text{red})$? Average the two scenarios:

$$\mathbb{P}(\text{red}) = \underbrace{\frac{3}{4} \cdot \frac{1}{2}}_{\text{Box 1}} + \underbrace{\frac{1}{4} \cdot \frac{1}{2}}_{\text{Box 2}} = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}.$$

Keep this number in mind: next we will *reverse* the question — “given a red token, which box was it probably from?” — and this $\frac{1}{2}$ becomes the denominator.

Independence: when information tells you nothing

Definition

A and B are **independent** if knowing one tells you nothing about the other:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A), \quad \text{equivalently} \quad \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Independence is what lets us multiply probabilities of separate trials — the coin does not remember its last flip. It is an assumption we make constantly, and it is usually an *approximation*.

Independent is not the same as disjoint

Common pitfall

Independent and *disjoint* are almost opposites. Disjoint events cannot both happen, so learning that one occurred tells you the other did *not* — that is maximal *dependence*. Independence means the events carry *no* information about each other.

Students conflate these two constantly. If in doubt, go back to the definition: independence is about $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, not about the events being mutually exclusive.

Why independence matters in modeling

Link to cognitive science

Independence assumptions are what make cognitive and machine-learning models computable. The “naive Bayes” classifier assumes features are independent given the category; many models of memory and categorisation assume trials are independent.

These assumptions are rarely exactly true, but they are often good enough — and a large part of the modeler’s craft is knowing *when* they break down.

Bayes' theorem: updating a belief

Now the destination. Bayes' theorem flips a conditional probability around, which is exactly what “learning from evidence” requires.

Key idea

$$\mathbb{P}(H | E) = \frac{\mathbb{P}(E | H) \mathbb{P}(H)}{\mathbb{P}(E)} \propto \mathbb{P}(E | H) \mathbb{P}(H),$$

where H is a hypothesis and E is the evidence we observe.

The next slide names the four pieces — each has a job worth learning.

The four pieces of Bayes' theorem

- **Prior** $\mathbb{P}(H)$ — what we believed *before* the evidence.
- **Likelihood** $\mathbb{P}(E | H)$ — how well the hypothesis *predicts* the evidence.
- **Posterior** $\mathbb{P}(H | E)$ — our *updated* belief.
- **Evidence** $\mathbb{P}(E)$ — a normaliser, from the law of total probability:
$$\mathbb{P}(E) = \mathbb{P}(E | H)\mathbb{P}(H) + \mathbb{P}(E | H^c)\mathbb{P}(H^c).$$

Intuition

New belief $\propto$ (how well the hypothesis predicted what we saw) $\times$ (how much we believed it beforehand).

The classic medical test — the setup

Example

A disease affects 1% of people. A test correctly flags 99% of true cases (sensitivity), but also wrongly flags 5% of healthy people (false positives). A randomly chosen person tests positive.

The natural question — “they tested positive, so they probably have it, right?” — turns out to be badly wrong. Let us see why, first with the formula, then with an even clearer picture.

The medical test — the calculation

Write D for “has the disease” and $+$ for “tests positive”. First the denominator, the overall chance of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \times 0.01}_{\text{sick} = 0.0099} + \underbrace{0.05 \times 0.99}_{\text{healthy} = 0.0495} = 0.0594.$$

Now Bayes:

$$\mathbb{P}(D | +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%.$$

A positive result means only about a 1-in-6 chance of disease.

The medical test — why, in plain numbers

The cleanest way to see it: imagine 10,000 people.

- 100 are sick; the test catches 99 of them.
- 9,900 are healthy; the test wrongly flags 5%, that is 495 of them.
- So $99 + 495 = 594$ people test positive, but only 99 are actually sick.

$$\mathbb{P}(\text{sick} \mid +) = \frac{99}{99 + 495} = \frac{1}{6}.$$

Because the disease is *rare*, the many false positives from the huge healthy group swamp the few true positives.

This is a famous cognitive bias

Link to cognitive science

People — including doctors — systematically overestimate $\mathbb{P}(D \mid +)$ because they focus on the test's accuracy and neglect the low base rate $\mathbb{P}(D)$. This is the **base-rate fallacy**, one of the most robust findings in the psychology of judgment (Kahneman & Tversky).

The very same mathematics describes Bayesian models of perception: the “prior” is the brain's expectation, and the “likelihood” is the incoming sensory evidence.

A shortcut: Bayes in odds form

Sometimes it is quicker to work with *odds* and skip the denominator entirely.

Useful fact

$$\underbrace{\frac{\mathbb{P}(H | E)}{\mathbb{P}(H^c | E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(E | H)}{\mathbb{P}(E | H^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}}.$$

Evidence simply *multiplies* the odds. For the medical test: prior odds 1:99, likelihood ratio $0.99/0.05 \approx 19.8$, giving posterior odds about 1:5 — the same 1/6 as before.

Practice — try it

A jar of candies is either *ordinary* (30% red) or *special* (70% red); beforehand each is equally likely. You draw one candy and it is red.

1. What is $\mathbb{P}(\text{red})$ overall?
2. What is $\mathbb{P}(\text{special} \mid \text{red})$?
3. Redo part (2) using prior odds $\times$ likelihood ratio.

From outcomes to numbers: random variables

So far outcomes have been labels (heads, even, positive). Usually we want *numbers* we can average and plot.

Definition

A **random variable** X assigns a number to every outcome. It is *discrete* if it takes separate values $(0, 1, 2, \dots)$ and *continuous* if it can take any value in a range.

Examples: the number of correct trials out of 20 (discrete); a reaction time in milliseconds (continuous); a spike count (discrete); a voltage (continuous).

Where a random variable's probability “lives”

Two functions describe a random variable, and one more accumulates it.

Useful fact

Discrete — the *probability mass function* $p(x) = \mathbb{P}(X = x)$; the masses are ≥ 0 and sum to 1.

Continuous — the *probability density function* $f(x)$; areas under it give probabilities, and the total area is 1.

Either — the *cumulative distribution function* $F(x) = \mathbb{P}(X \leq x)$, rising from 0 to 1.

A mass function is a bar chart; a density is the smooth curve you get as the bars become infinitely thin.

A subtlety worth stressing

Common pitfall

For a continuous variable, $f(x)$ is a *density*, not a probability. It can be larger than 1, and the probability of any *single exact* value is zero. Only *intervals* (areas) carry probability.

So “the probability the reaction time is exactly 300.000... ms” is zero, but “between 290 and 310 ms” is a real, positive number — the area under the curve across that interval.

Practice — try it

A continuous variable has density $f(x) = 2x$ for $0 \leq x \leq 1$, and 0 elsewhere.

1. Check that this is a valid density.
2. Find $\mathbb{P}(X \leq 0.5)$.
3. Explain why $\mathbb{P}(X = 0.5) = 0$.

Four distributions cover most of what you will meet

A “distribution” is a named, reusable shape of randomness. We will meet four, each with its own story — learn the story, not just the formula.

- **Bernoulli** — a single yes/no trial.
- **Binomial** — counting successes across many trials.
- **Poisson** — counts of rare events (e.g. spikes).
- **Gaussian** — the bell curve of noise and averages.

Bernoulli and Binomial

Bernoulli(p) is one yes/no trial: success with probability p . It is the atom of all binary data — one coin flip, one correct/incorrect response.

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Binomial(n, p) counts the successes in n independent Bernoulli trials:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E}[X] = np.$$

So a Binomial is just a *sum* of n Bernoullis.

Guessing on a multiple-choice test

A student guesses on 10 four-option questions, so $p = \frac{1}{4}$ on each.

The probability of exactly 3 correct:

$$\mathbb{P}(X = 3) = \binom{10}{3} (0.25)^3 (0.75)^7 \approx 0.250.$$

The expected number correct is $\mathbb{E}[X] = np = 10 \times 0.25 = 2.5$. This is why pure guessing on a 10-question quiz typically yields about 2–3 right.

Poisson — counts of rare events

When events happen independently at a steady average rate, the number in a fixed window follows a **Poisson**(λ), where λ is *both* the mean and the variance:

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \mathbb{E}[X] = \text{Var}(X) = \lambda.$$

Link to cognitive science

This is the standard first model of **neural spike counts**. Its signature, “variance = mean”, is used all the time to check whether real spike data are more or less variable than this baseline.

Example

A neuron fires on average $\lambda = 5$ spikes per second. What is the probability of observing exactly 3 spikes in a given second?

$$\mathbb{P}(X = 3) = \frac{5^3 e^{-5}}{3!} = \frac{125 \times 0.006738}{6} \approx 0.140.$$

So even though the *average* is 5, seeing only 3 spikes in a particular second is entirely ordinary — about a 14% chance. Spike counts are genuinely noisy.

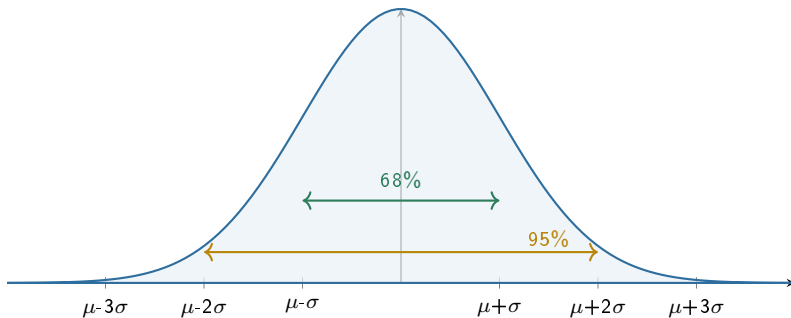
The Gaussian: the bell curve

The **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$ is the continuous workhorse. It is symmetric and bell-shaped, centred at the mean μ , with its width set by the standard deviation σ :

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The next slide shows the one fact about it that lets you reason without ever doing an integral.

The 68–95–99.7 rule



About 68% of the probability lies within $\pm 1\sigma$ of the mean, 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$ — for every Normal distribution.

Reading off Normal probabilities

To use the rule for any Normal, we *standardise*: measure how many standard deviations a value is from the mean, $z = \frac{x - \mu}{\sigma}$.

Example

Adult heights are roughly $\mathcal{N}(\mu = 170, \sigma = 10)$ cm. What fraction of people are taller than 185 cm?

Standardise: $z = \frac{185 - 170}{10} = 1.5$. From a z-table (or the rule), $\mathbb{P}(Z > 1.5) \approx 0.067$ — about 6.7% of people exceed 185 cm.

The four distributions at a glance

Distribution	Models	Support	Mean	Variance
Bernoulli(p)	one yes/no trial	$\{0, 1\}$	p	$p(1 - p)$
Binomial(n, p)	# successes in n	$\{0, \dots, n\}$	np	$np(1 - p)$
Poisson(λ)	counts of rare events	$\{0, 1, 2, \dots\}$	λ	λ
Normal(μ, σ^2)	noise / averages	$\mathbb{R}$	μ	σ^2

Two links: a Binomial is a sum of Bernoullis; and for large n a Binomial looks Normal (a preview of the Central Limit Theorem).

Why the bell curve is everywhere

Link to cognitive science

Whenever a quantity is the sum of many small, independent influences, its distribution is approximately Normal — regardless of the details. This is the **Central Limit Theorem**, which we will make intuitive next session.

It explains why test scores, measurement errors, and averaged neural signals all look bell-shaped, and why Gaussian noise is the default assumption in almost every model of perception and data analysis.

Practice — try it

1. A subject responds correctly on 70% of trials, independently. Over 8 trials, what is the expected number correct, and $\mathbb{P}(\text{all 8 correct})$?
2. A neuron fires at $\lambda = 2$ spikes per window. What is $\mathbb{P}(X = 0)$?
3. Reaction times are $\mathcal{N}(500, 50^2)$ ms. Roughly what fraction fall between 450 and 550 ms?

Session 1 — what we built

- Probabilities are numbers in $[0, 1]$ obeying the complement and addition rules.
- **Conditional probability** formalises learning from information; *independence* is its absence.
- **Bayes' theorem** turns evidence into updated belief — the engine of Bayesian cognition and the base-rate fallacy.
- **Random variables** and four *distributions* describe binary trials, counts, rare events, and bell-shaped noise.

Next session: from single quantities to *data* — summarising it and inferring from it.

Session 2 — Statistics & Inference

What today is about

Last session we studied *single* uncertain quantities. Today we turn to *data*: many observations at once, and what we can learn from them.

- Summarise data with a few numbers: *mean, variance, covariance*.
- Understand why averaging works: the *Law of Large Numbers* and the *Central Limit Theorem*.
- Infer from a sample to a population: *standard errors, confidence intervals, hypothesis tests*.
- Build our first real model: *linear regression*.

The thread running through everything

Population vs. sample

There is a fixed but unknown truth about the whole *population* (for example, the true average reaction time of all humans). We only ever see a limited *sample*. Statistics is the disciplined art of guessing the population from the sample — and honestly reporting how uncertain that guess is.

Keep asking, on every slide: is this a statement about the population, or about our particular sample?

The centre of a distribution: expectation

The first thing we want to know about data is where it is centred.

Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

Picture the distribution as a pile of mass on the number line: the expectation is its *balance point*, where a finger would keep it level. It need not be a value the variable can actually take — a fair die has mean 3.5.

Expectation adds up nicely

The property that makes expectation so useful is *linearity* — and it holds always, even when variables influence each other.

Useful fact

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

Example

A fair die has $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + \cdots + 6) = 3.5$. A game paying $Y = 2X + 1$ points has $\mathbb{E}[Y] = 2(3.5) + 1 = 8$ — no need to re-average the six payouts.

How spread out is the data? Variance

The second thing we want is how much the data spreads around its centre.

Definition

The **variance** is the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

We square the deviations so positive and negative ones do not cancel, and to penalise large departures more. The square root then brings us back to sensible units — milliseconds, not milliseconds-squared.

Three variance rules worth memorising

Useful fact

Computational formula: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ — usually the fastest by hand.

Shift and scale: $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (adding a constant does not change spread).

Sum: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

That last term — the covariance — is exactly the correction for when X and Y move together. It vanishes only when they are uncorrelated.

Example

Find the variance of a fair die.

Using the computational formula, we need the mean of the square:

$$\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.17.$$

Then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 15.17 - 3.5^2 = 2.92, \quad \text{SD}(X) \approx 1.71.$$

So a die roll typically lands about 1.7 away from its mean of 3.5.

How two variables move together

Often we care about the relationship between two quantities — study time and test score, or the firing of two neurons.

Definition

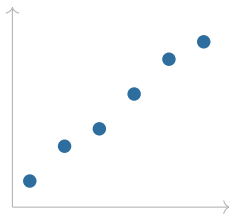
The **covariance** measures whether they tend to be large or small together:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)],$$

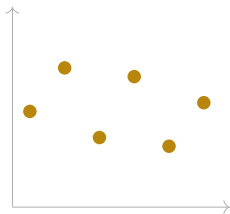
and the **correlation** rescales it to lie in $[-1, 1]$: $\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$

Covariance is positive when the two are usually on the same side of their means, negative when one is high while the other is low.

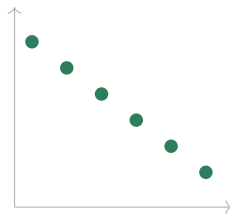
What correlation looks like



$$\rho \approx +0.9$$



$$\rho \approx 0$$



$$\rho \approx -0.9$$

Correlation measures the strength and direction of a *linear* trend: +1 a perfect rising line, 0 no linear trend, -1 a perfect falling line.

Covariance is the bridge to linear algebra

Key idea

When we have many variables at once, all their pairwise covariances are collected into a **covariance matrix**. That single object is the heart of Principal Component Analysis in Session 4. Keep the word in mind — it returns.

This is one of the places where the two halves of the course quietly become the same subject: a statistical quantity (covariance) turns into a geometric object (a matrix we can rotate and decompose).

Example

Five students' weekly study hours $x = (1, 2, 3, 4, 5)$ and quiz scores $y = (2, 4, 5, 4, 5)$.

The means are $\bar{x} = 3$, $\bar{y} = 4$. Summing the deviations gives

$$\sum (x - \bar{x})(y - \bar{y}) = 6, \quad \sum (x - \bar{x})^2 = 10, \quad \sum (y - \bar{y})^2 = 6.$$

So the correlation is

$$\rho = \frac{6}{\sqrt{10 \times 6}} = \frac{6}{\sqrt{60}} \approx 0.77 :$$

a strong positive association. (We reuse these same sums for the regression line at the end of today.)

A correlation warning

Common pitfall

$\rho = 0$ means no *linear* relationship — not “no relationship”. Points on a symmetric parabola $y = x^2$ have zero correlation despite being perfectly predictable. And correlation is never, by itself, evidence of *causation*.

In a field full of observational data, this is worth repeating often: a strong correlation is an invitation to investigate, not a conclusion about cause.

Practice — try it

A variable X takes the values $\{2, 4, 6\}$, each with probability $\frac{1}{3}$.

1. Find $\mathbb{E}[X]$.
2. Find $\text{Var}(X)$ using the computational formula.
3. If $Y = 3X$, what is $\text{SD}(Y)$?

Two theorems behind all of statistics

Why can we trust an average of noisy measurements? Two results answer this.

Key idea

Law of Large Numbers: as the sample size n grows, the sample mean $\bar{X}_n$ converges to the true mean μ . *Averages settle down.*

Central Limit Theorem: for large n , the sample mean is approximately Normal, $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$, *whatever the shape of the original data. Averages become bell-shaped.*

Why the Central Limit Theorem is remarkable

The Law of Large Numbers is intuitive: more data, better estimate. The Central Limit Theorem says something stronger and more surprising.

Even if individual observations are wildly non-Normal — a skewed reaction time, a 0/1 correct response — their *average* over many trials traces out a smooth Normal curve, and its spread shrinks in a precise way, like $\sigma/\sqrt{n}$.

This is why the Normal distribution is unavoidable, and why error bars have the shape they do.

Averaging as noise reduction

Link to cognitive science

The Central Limit Theorem underwrites the everyday practice of *trial averaging* in EEG and fMRI. Averaging n noisy trials cuts the noise standard deviation by a factor of $\sqrt{n}$, pulling a buried signal out of the noise.

It also explains why so many measured traits look Gaussian: each is the aggregate of many small independent contributions — many neurons, many trials, many genes and experiences behind a single behavioural score.

The $\sqrt{n}$ law of averaging

Useful fact

If $X_1, \dots, X_n$ are independent with mean μ and standard deviation σ , the sample mean $\bar{X}$ has

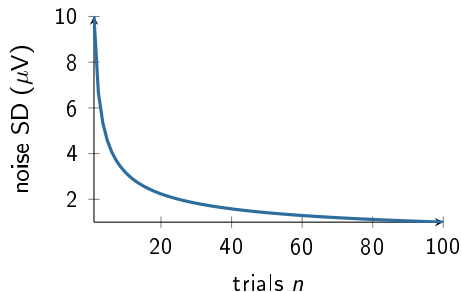
$$\mathbb{E}[\bar{X}] = \mu, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The key consequence is the *diminishing return*: to *halve* the uncertainty of an estimate you need *four times* as much data; to cut it tenfold, a hundredfold. This is why large studies are expensive.

The $\sqrt{n}$ law — picture and example

Example

Single-trial EEG responses have noise SD $\sigma = 10 \mu\text{V}$. Averaging 25 trials gives noise $10/\sqrt{25} = 2 \mu\text{V}$. To reach $1 \mu\text{V}$ we need $10/\sqrt{n} = 1$, i.e. $n = 100$ — four times as many trials.



Estimating what we cannot see

We almost never observe a whole population; we observe a sample and use it to guess.

Definition

A **parameter** is a fixed unknown describing the population (e.g. the true mean μ). An **estimator** is a recipe applied to the sample to approximate it (e.g. the sample mean $\bar{X}$). It is **unbiased** if it is right *on average* across many samples: $\mathbb{E}[\hat{\theta}] = \theta$.

Why we divide by $n - 1$

Common pitfall

The sample variance divides by $n - 1$, not n :

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Using the sample mean $\bar{x}$ (instead of the unknown true mean) makes the deviations come out slightly too small. Dividing by $n - 1$ rather than n corrects exactly for this, giving an unbiased estimate. Students should know both the rule and the reason.

Standard deviation vs. standard error

This distinction is subtle and constantly misused — so it gets its own slide.

Common pitfall

The **standard deviation** s describes the spread of the raw data; it does *not* shrink with more data. The **standard error** $SE = s/\sqrt{n}$ describes the precision of our *estimate of the mean*; it *does* shrink as n grows.

When a figure shows error bars, always ask which one they are: the spread of individuals, or the uncertainty in the mean? They tell very different stories.

Reporting uncertainty: confidence intervals

A single estimate hides its own uncertainty. A confidence interval reports a *range* of plausible values.

Definition

A 95% confidence interval for the mean is

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

The width $1.96 \sigma / \sqrt{n}$ comes straight from the Central Limit Theorem and the 95% mark of the Normal curve. The recipe is built so that, across many repeated samples, 95% of the intervals it produces contain the true μ .

Useful fact

In practice we estimate σ by the sample SD s . This extra uncertainty means the standardised mean follows *Student's t -distribution* with $n - 1$ degrees of freedom — like the Normal, but with heavier tails.

For small samples the 95% multiplier is larger than 1.96 (about 2.26 when $n = 10$), and it shrinks toward 1.96 as n grows. A practical rule of thumb: once $n \gtrsim 30$, the difference hardly matters.

Confidence interval — worked example

Example

A test with known population SD $\sigma = 15$ is given to $n = 25$ people; the sample mean is $\bar{x} = 105$.

The standard error is $SE = 15/\sqrt{25} = 15/5 = 3$, so the 95% interval is

$$105 \pm 1.96 \times 3 = 105 \pm 5.88, \quad \text{i.e. about } [99.1, 110.9].$$

We would report: our best estimate of the population mean is 105, and values from roughly 99 to 111 are plausible given this sample.

What “95% confidence” really means

Common pitfall

A 95% confidence interval does *not* mean “there is a 95% probability the true mean lies in *this* interval”. In the classical framework μ is fixed; it is the interval that is random.

The correct reading is about the *procedure*: if we repeated the whole experiment many times, 95% of the intervals we build this way would capture the truth. Getting this subtlety right is part of being a careful analyst.

Practice — try it

1. A sample of $n = 100$ has mean $\bar{x} = 52$ with known $\sigma = 10$. Give a 95% confidence interval for μ .
2. Testing $H_0 : \mu = 50$ against $\mu \neq 50$, what is the z-statistic, and the rough conclusion?
3. A treatment raises scores by 2 points, with within-group SD 10. What is Cohen's d , and is it small, medium, or large?

The logic of a hypothesis test

The dominant — and most criticised — ritual of statistics. Students must be able to both perform it and understand its limits.

Four steps

- (1) State a **null hypothesis** H_0 (“nothing is going on”) and an alternative H_1 .
- (2) Choose a **test statistic** that would be extreme if H_1 were true.
- (3) Compute the **p -value**: assuming H_0 , the probability of a result at least this extreme.
- (4) If $p < \alpha$ (often 0.05), **reject** H_0 .

Hypothesis test — worked example

Example

A coin lands heads 60 times in 100 flips. Is it fair?

Under $H_0 : p = 0.5$, the number of heads has mean $np = 50$ and standard deviation $\sqrt{np(1-p)} = \sqrt{25} = 5$. The standardised statistic is

$$z = \frac{60 - 50}{5} = 2.0.$$

A result this extreme (either tail) has probability $\mathbb{P}(|Z| \geq 2.0) \approx 0.046$. Since $0.046 < 0.05$, we reject H_0 : weak but statistically significant evidence of bias.

Definition

A **Type I error** is rejecting a true H_0 (a false positive), with probability α . A **Type II error** is failing to reject a false H_0 (a false negative), with probability β . The **power** of a test is $1 - \beta$: its chance of detecting a real effect.

There is always a trade-off: making α very small (few false positives) tends to lower power (more missed real effects). Choosing that balance is a design decision, not a mathematical one.

The workhorse tests

All follow the same four-step logic; they differ only in the statistic used.

- **t -test** — compares means (one sample, or two groups).
- **χ^2 test** — compares observed vs. expected counts in categories.
- **ANOVA** (F -test) — compares means across three or more groups.

Each carries assumptions worth stating out loud: roughly, independence of observations and approximate Normality of the relevant averages.

Significant is not the same as big

A real effect the test misses

Two groups of 25 score means 105 vs. 100, pooled SD 10. Then $SE = 10\sqrt{2/25} \approx 2.83$, $t \approx 1.77$, giving $p \approx 0.08$ — *not* significant. Yet the effect size is $d = \frac{5}{10} = 0.5$, a solidly “medium” effect. The study is simply underpowered.

The lesson: always report an *effect size* (like Cohen's d or ρ) alongside the p -value.

The p -value and the replication crisis

Link to cognitive science

The p -value is routinely misunderstood. It is *not* the probability that H_0 is true, nor that the result was “due to chance”. Misuse of $p < 0.05$ — through multiple comparisons, optional stopping, and publication bias — is a central driver of the **replication crisis** that has reshaped psychology and cognitive neuroscience since the 2010s.

Reporting effect sizes and confidence intervals, and pre-registering analyses, are part of the modern remedy.

From correlation to prediction: regression

Correlation measures the strength of a linear relationship. Regression draws the actual line and lets us *predict*.

Definition

Simple linear regression fits $\hat{y} = a + bx$ by **least squares** — choosing a, b to minimise the total squared vertical error $\sum (y_i - \hat{y}_i)^2$. The solution is

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad a = \bar{y} - b\bar{x}.$$

What least squares is doing

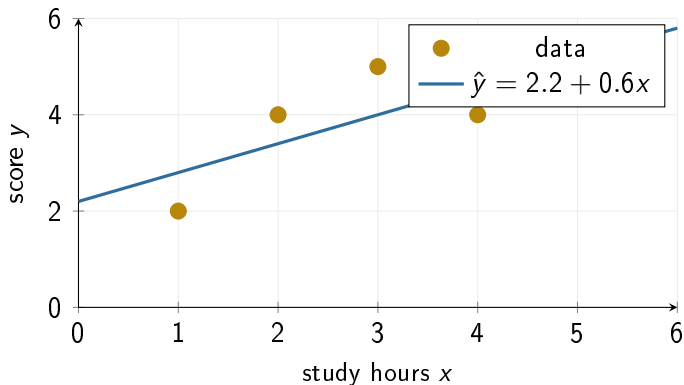
Intuition

We look for the straight line that passes as close as possible to all the points, where “close” is measured by the squared vertical gaps between each point and the line. The slope turns out to be just the covariance of x and y , rescaled by the spread of x — the same covariance from an hour ago.

Two facts that make good checks

The residuals always sum to zero and are uncorrelated with x ; and the line always passes through the mean point $(\bar{x}, \bar{y})$.

The regression line, drawn



The fitted line through the study-hours data. Each extra hour of study predicts about 0.6 more points.

Example

Using the study-hours data, with $\sum(x - \bar{x})(y - \bar{y}) = 6$ and $\sum(x - \bar{x})^2 = 10$.

$$b = \frac{6}{10} = 0.6, \quad a = 4 - 0.6 \times 3 = 2.2, \quad \hat{y} = 2.2 + 0.6x.$$

The fraction of variance the line explains is $R^2 = \rho^2 \approx 0.77^2 \approx 0.60$: the line accounts for about 60% of the variation in scores.

Regression is the atom of modeling

Link to cognitive science

Extend regression to many predictors and it becomes *multiple regression*; add a nonlinear “link” and it becomes the *Generalized Linear Model* behind psychometric functions and neural encoding models (Session 5); view it as “fit parameters to minimise error” and it is exactly the objective of supervised machine learning.

And geometrically, least squares is a *projection* — which is where linear algebra picks up the story next session.

Practice — try it

Four points: $x = (0, 1, 2, 3)$, $y = (1, 1, 2, 4)$.

1. Find $\bar{x}$ and $\bar{y}$.
2. Compute $\sum(x - \bar{x})(y - \bar{y})$ and $\sum(x - \bar{x})^2$.
3. Give the regression line $\hat{y} = a + bx$.

- Summaries: the *mean* (centre), *variance/SD* (spread), and *covariance/correlation* (co-movement).
- The *LLN* and *CLT*: averages converge and become Normal with spread $\sigma/\sqrt{n}$ — the basis of standard errors and confidence intervals.
- *Hypothesis testing*: *p*-values, the errors, and why significance is not importance.
- *Regression*: a least-squares line, reusing covariance and pointing ahead to the GLM and machine learning.

Next: linear algebra — vectors, matrices, and geometry.

Session 3 — Linear Algebra I

What today is about

Linear algebra is the mathematics of *data in many dimensions* and of *linear transformations*. It is the language of almost every model you will meet later.

- A *vector*: a point, an arrow, or a list of features.
- The *dot product*: similarity, length, angle, and projection.
- A *matrix*, read two ways: a data table *and* a transformation.
- *Linear systems*, the *inverse*, and *rank* — the true dimensionality of data.

The single operation everything is built from

Linear combination

Take some vectors, scale each by a number, and add them up:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Almost every concept today — span, basis, matrix multiplication, and next session's PCA — is really a statement about linear combinations.

If this operation feels natural by the end of the session, everything else follows.

A vector, read three ways

Definition

A **vector** $v \in \mathbb{R}^n$ is an ordered list of numbers. We can read it as a *point* at those coordinates, an *arrow* from the origin to that point, or a *list of features* describing one object.

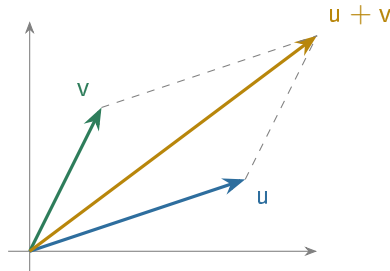
All three readings matter. “Arrow” gives us geometry (length, direction, angle). “Point” lets us picture a whole dataset as a cloud of points. “List of features” is how real data arrives: a participant is a vector of (age, reaction time, accuracy, ...); an image is a vector of pixel values.

Adding and scaling vectors

We combine vectors in just two ways:

- **Add** them componentwise — geometrically, place them tip-to-tail.
- **Scale** them by a number — stretch the arrow, or flip it if the number is negative.

Every richer operation is built from these two.



The dot product: one operation, three meanings

Definition

The **dot product** multiplies matching components and adds:

$$u \cdot v = \sum_i u_i v_i = \|u\| \|v\| \cos \theta,$$

where θ is the angle between the arrows.

From it we also read off *length* $\|v\| = \sqrt{v \cdot v}$ and *distance* $\|u - v\|$. So a single number quietly encodes similarity, size, and angle all at once.

The dot product is a similarity score

Intuition

Vectors pointing the same way have a large positive dot product; perpendicular vectors have dot product zero (“nothing in common”); opposite vectors have a large negative one.

Divide the dot product by the two lengths and you get $\cos \theta$ — the **cosine similarity** used everywhere to compare feature vectors, because it ignores their overall magnitude and keeps only their direction.

Dot product — worked example

Example

Let $u = (1, 2)$ and $v = (3, 1)$.

The dot product is $u \cdot v = 1 \cdot 3 + 2 \cdot 1 = 5$. The lengths are $\|u\| = \sqrt{5}$ and $\|v\| = \sqrt{10}$. So

$$\cos \theta = \frac{5}{\sqrt{5} \sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \theta = 45^\circ.$$

The two arrows sit at a 45° angle — a moderate positive similarity.

A few facts about the dot product

Useful fact

Cauchy–Schwarz: $|u \cdot v| \leq \|u\| \|v\|$ — which is why cosine similarity always lies in $[-1, 1]$.

Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$ — a detour is never shorter than the direct route.

Unit vector: $\hat{u} = u / \|u\|$ keeps only the direction.

Orthogonality: the geometry of “unrelated”

Key idea

Two vectors are *perpendicular (orthogonal)* exactly when their dot product is zero. This is the geometric meaning of “no linear relationship” — and it is the same idea as zero correlation once data are mean-centred.

It connects directly to last session’s regression: the best-fit line is the one whose residual vector is orthogonal to the predictors. That is the bridge we make concrete next.

Projection: the geometry of least squares

Useful fact

The projection of b onto the direction of a is

$$\text{proj}_a b = \frac{a \cdot b}{a \cdot a} a,$$

and the leftover (the *residual*) is orthogonal to a .

The projection is the “shadow” of b along the direction of a ; the residual is the part of b that a simply cannot explain. Fitting a regression line is exactly projecting the data onto the space of predictors.

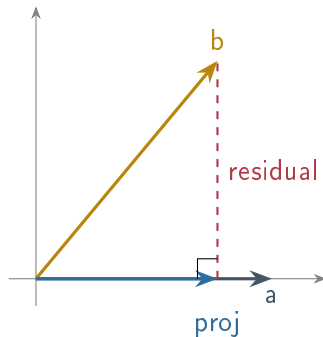
Projection — picture and example

Example

Project $b = (2, 3)$ onto $a = (1, 1)$:

$$\frac{a \cdot b}{a \cdot a} a = \frac{5}{2}(1, 1) = (2.5, 2.5).$$

The residual $(-0.5, 0.5)$ satisfies $a \cdot (-0.5, 0.5) = 0$ — perpendicular, as it must be.



Link to cognitive science

Representing stimuli, words, or neural population states as vectors — and comparing them by dot product or cosine similarity — is the backbone of *representational similarity analysis* in neuroscience and of *word embeddings* in computational linguistics.

“King — man + woman $\approx$ queen” is literally vector arithmetic. A population of neurons firing is a single point in a high-dimensional space, and how two conditions differ is just a distance between two such points.

Practice — vectors and the dot product

Practice — try it

Let $u = (3, 4)$ and $v = (4, -3)$.

1. Find $\|u\|$.
2. Compute $u \cdot v$. Are the two vectors orthogonal?
3. Give the unit vector in the direction of u .

A matrix has two readings — keep both

Definition

An $m \times n$ **matrix** is a rectangular grid of numbers. Read it as a *data table* (rows are observations, columns are features) *and* as a *linear transformation* $x \mapsto Ax$ that moves every vector to a new place.

The data-table reading is how we *store* observations. The transformation reading is how models *act* on them — for example, a layer of a neural network multiplies its input by a weight matrix.

A matrix transforms space

Intuition

A matrix sends the whole grid of vectors to a new grid, keeping the origin fixed and straight lines straight. The *columns* tell you exactly where the basis vectors land: the first column is where $(1, 0)$ goes, the second where $(0, 1)$ goes. Everything else follows by linear combination.

Rotations, reflections, scalings, and shears are all just different matrices — different instructions for where the two axis arrows should end up.

Matrix times vector = combination of columns

The product Ax is a linear combination of the columns of A , weighted by the entries of x :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} a \\ c \end{pmatrix} + y \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

A rotation

$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ sends $(1, 0)$ to $(0, 1)$: the arrow pointing right becomes the arrow pointing up — a quarter turn. The columns of R are where the two axis vectors land.

Multiplying two matrices = composing transformations

The product AB means “do B , then A ”. Entry (i,j) is the dot product of row i of A with column j of B .

Example

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

Right-multiplying by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ swapped the *columns* of the first matrix.

The algebra of matrices — and one warning

Useful fact

Matrix multiplication is *associative* and *distributive*; $(AB)^T = B^T A^T$; and $AI = A$.

Common pitfall

Matrix multiplication is **not commutative**: in general $AB \neq BA$. Order encodes the order of operations — “rotate then stretch” differs from “stretch then rotate”. Also, inner dimensions must match: an $m \times n$ times an $n \times p$ gives $m \times p$.

Practice — try it

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ (a horizontal shear) and $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

1. Compute Ax .
2. Compute AA (apply the shear twice).
3. Does A change area? Find $\det A$.

A system of equations is one matrix equation

Definition

The system $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$ is written compactly as $Ax = b$. Solving it means finding the vector x that A maps to b .

Two ways to picture it. *Row picture*: each equation is a line, and the solution is where the lines cross. *Column picture*: which combination of A 's columns produces b ? The column view is the one that scales up to many dimensions.

Undoing a transformation: the inverse

Definition

The **inverse** A^{-1} undoes A : $A^{-1}A = I$. For a 2×2 matrix, with $\det A = ad - bc$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (\text{when } \det \neq 0),$$

and then the system has the unique solution $x = A^{-1}b$.

The determinant, and when things break

Intuition

The determinant measures how much the transformation *scales area* (or volume). If $\det A = 0$, the transformation squashes space flat — it collapses a whole dimension — so information is lost and it cannot be undone.

A zero determinant is the algebraic warning sign that a system may have no solution, or infinitely many. A nonzero determinant guarantees exactly one.

Example

Invert $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

The determinant is $\det A = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0$, so the inverse exists:

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}.$$

You can check that $AA^{-1} = I$.

Solving larger systems by elimination

Gaussian elimination: use one equation to cancel a variable from the others, repeat, then back-substitute. It is the 2×2 idea, scaled up.

Example

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

Removing x from the last two equations gives $3y + z = 9$ and $y - 2z = -4$. Substituting $z = 9 - 3y$ into the second: $7y = 14$, so $y = 2$, then $z = 3$ and $x = 1$. The solution is $(x, y, z) = (1, 2, 3)$.

Definition

Vectors are **linearly independent** if none is a combination of the others (no redundancy). The **span** is all combinations you can build; a **basis** is a minimal independent set that spans the space, and its size is the **dimension**. The **rank** of a matrix is its number of independent columns.

Rank is the honest count of how many dimensions the data actually occupies — which can be far fewer than the number of columns.

Quick tests, and an example

Useful fact

Two vectors are dependent exactly when one is a multiple of the other. Square columns are independent exactly when $\det \neq 0$. And any more than n vectors in $\mathbb{R}^n$ are automatically dependent.

Example

For $\begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \end{pmatrix}$: column 2 is $3 \times$ column 1 (no new direction), but column 3 is not a multiple of column 1. So there are two independent directions: $\text{rank} = 2$.

Why rank matters in neuroscience

Link to cognitive science

This is exactly what we mean by the *dimensionality* of neural data. A recording of 100 neurons lives in $\mathbb{R}^{100}$, but their activity during a task often varies along only a handful of independent directions — the data have low rank.

Finding those few directions is *dimensionality reduction*, and the principled way to do it is Principal Component Analysis — exactly where next session begins.

Practice — rank and independence

Practice — try it

1. Are $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ independent?
2. Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ independent?
3. Can three vectors in $\mathbb{R}^2$ ever be independent?

Session 3 — what we built

- Vectors are points/arrows/features, combined by *linear combination*.
- The *dot product* gives length, distance, angle, and similarity; *orthogonality* is “unrelated”; *projection* is least squares.
- A *matrix* is both a data table and a transformation; multiplication composes maps, and is not commutative.
- The *inverse* exists when $\det \neq 0$; *rank* is the true dimensionality of data.

Next: the special directions of a matrix — eigenvectors — and their flagship application, PCA.

Session 4 — Linear Algebra II

What today is about — the payoff session

Today the two halves of the course meet.

- *Eigenvectors* and *eigenvalues*: the special directions of a transformation.
- **Principal Component Analysis**: the flagship tool for reducing and visualising high-dimensional data.
- The *SVD*: the general engine behind PCA.
- *Dynamics*: how eigenvalues decide whether a system settles, oscillates, or explodes.

Key idea

The **covariance matrix** from Session 2 (a statistical object) read through its **eigenvectors** (a linear-algebra tool) *is* Principal Component Analysis. That single sentence is the destination of today.

So keep the covariance matrix in mind as we build the eigenvalue machinery — we are assembling exactly the pieces PCA needs.

The special directions of a matrix

Most vectors get knocked off their direction when a matrix acts on them. A few special ones do not.

Definition

A nonzero vector v is an **eigenvector** of A if A only *stretches* it, without changing its direction:

$$Av = \lambda v,$$

where the number λ is the **eigenvalue** — the stretch factor.

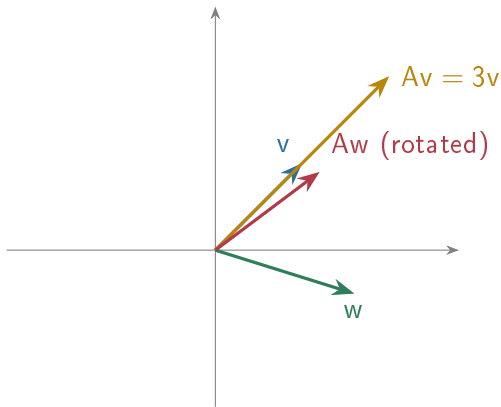
Eigenvectors are the “grain of the wood”

Intuition

Think of the transformation as flowing space around. Almost every arrow gets rotated as well as stretched. But along an eigenvector, the transformation is pure stretch: the arrow stays on its own line, only getting longer, shorter, or flipped.

Along these special directions a complicated matrix behaves like plain multiplication by a number. That simplicity is what makes eigenvectors so powerful.

Eigenvectors — the picture



Along the eigenvector v the matrix only stretches (here by 3). A generic vector w gets rotated off its line — it is not an eigenvector.

How to find them

To find eigenvalues we solve the **characteristic equation** $\det(A - \lambda I) = 0$. Each root λ then gives its eigenvectors by solving $(A - \lambda I)v = 0$.

Example

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = 0 \Rightarrow \lambda = 3 \text{ or } 1.$$

The eigenvectors are $(1, 1)$ for $\lambda = 3$ and $(1, -1)$ for $\lambda = 1$ — and they are *orthogonal*.

Two free checks on your eigenvalues

Useful fact

For any square matrix:

$$\sum_i \lambda_i = \text{tr}(A) \text{ (sum of the diagonal),} \quad \prod_i \lambda_i = \det(A).$$

For the example above: $\lambda_1 + \lambda_2 = 3 + 1 = 4 = \text{tr}(A)$, and $\lambda_1 \lambda_2 = 3 \cdot 1 = 3 = \det(A)$.
Always use these to catch arithmetic slips before going any further.

Why eigenvalues make repeated action easy

Diagonalization

Collect the eigenvectors as columns of P and the eigenvalues on the diagonal of D ; then $A = PDP^{-1}$, and powers become trivial:

$$A^k = PD^kP^{-1}, \quad D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

In the eigenvector coordinates the matrix is just a set of independent stretches. This is why the long-run behaviour of A^k is governed entirely by the eigenvalues — exactly what we need for dynamics later today.

Symmetric matrices are especially nice

Key idea

If $A = A^T$ (symmetric — as every covariance matrix is), then its eigenvalues are real and its eigenvectors are mutually orthogonal. The transformation is then just a set of independent stretches along perpendicular axes.

A symmetric matrix is *positive-definite* when all its eigenvalues are positive. Covariance matrices are positive-semidefinite — their eigenvalues are never negative, because they represent amounts of variance. This clean structure is exactly what makes PCA work.

Practice — try it

Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$ (already diagonal).

1. What are its eigenvalues and eigenvectors?
2. Check the trace and the determinant against your eigenvalues.
3. What does A do to the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$?

The problem PCA solves

High-dimensional data are hard to see and often redundant. Because many measured variables are correlated, the cloud of data points really lies close to a lower-dimensional subspace.

PCA finds the few directions that capture the most variation, so we can describe, visualise, and compress the data with little loss. It answers: *along which directions does the data actually vary?*

Key idea

The principal components are the **eigenvectors of the covariance matrix**, and each eigenvalue is the **amount of variance** along that direction. The first component is the direction of greatest spread; the next is the greatest remaining spread orthogonal to it; and so on.

Everything from the first half of today — eigenvectors, symmetric matrices, orthogonality — comes together in this one idea.

The PCA recipe

1. **Centre** the data: subtract each variable's mean, so the cloud sits at the origin.
2. Form the **covariance matrix** C of the centred data.
3. Compute the **eigenvectors and eigenvalues** of C .
4. Order the eigenvectors by eigenvalue, largest first: these are principal components $1, 2, 3, \dots$
5. **Project** the data onto the top few components to reduce dimension.

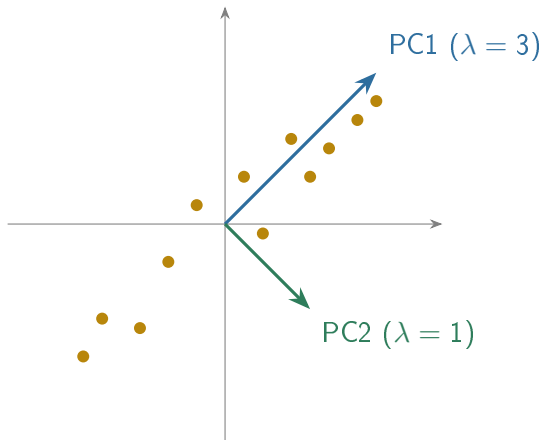
What PCA is doing, geometrically

Intuition

Picture an elongated, tilted cloud of points. PCA finds the long axis of the cloud (the direction of most variance — the first component) and the short axis perpendicular to it (the second).

Rotating onto these axes “straightens” the cloud so its directions become uncorrelated. Keeping only the long axis is dimensionality reduction: we describe each point by its position along the direction that matters most.

PCA — the picture



PC1 follows the long axis of the cloud (most variance); PC2 is perpendicular, capturing the little that remains.

Example

Two mean-centred variables have covariance matrix $C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

We already found its eigenpairs: $\lambda_1 = 3$ along $(1, 1)$, and $\lambda_2 = 1$ along $(1, -1)$. So the first principal component points along $(1, 1)$ (the two variables rise together) and carries variance 3; the second carries variance 1. The fraction of variance explained by the first component is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{4} = 75\%.$$

Useful fact

The **total variance** is the sum of the eigenvalues, equal to the trace of C . The **proportion explained** by component j is $\lambda_j / \sum_i \lambda_i$; plotting these in decreasing order gives the **scree plot**, whose “elbow” suggests how many components to keep. Because C is a covariance matrix, all $\lambda_i \geq 0$.

A caution about PCA

Common pitfall

PCA is a *linear* method and is sensitive to *scaling*. A variable measured in large units (say reaction time in milliseconds) can dominate the components purely because of its units. Standardising the variables, or using the correlation matrix, is often necessary.

Also remember: PCA finds directions of *variance*, which are not always the directions that matter for a task. Variance is not the same as relevance.

Link to cognitive science

PCA reduces recordings of hundreds of neurons to a few “latent” dimensions along which population activity evolves (neural state-space analysis); it compresses fMRI voxels and EEG channels; it extracts the principal components of face images (“eigenfaces”); and it summarises questionnaire batteries into a few underlying factors.

Whenever someone says a system is “effectively low-dimensional”, PCA is usually how they found out.

Practice — try it

After PCA on some neural data, the covariance has eigenvalues $\lambda = (6, 3, 1)$ along three orthogonal components.

1. What is the total variance?
2. What fraction does the first component explain?
3. How many components are needed to keep at least 85% of the variance?

The SVD: the general engine behind PCA

Definition

The **Singular Value Decomposition** writes *any* matrix as $X = U\Sigma V^T$: a rotation, a set of non-negative stretches (the *singular values* in Σ), and another rotation.

The idea to carry away: *every* linear transformation is a rotation, then a scaling along perpendicular axes, then another rotation. PCA is just the SVD applied to the centred data matrix, and the singular values squared are the variances. SVD also handles non-square data, which PCA's covariance route cannot.

Eigenvalues also predict the future

Many models describe how a state x evolves, either step by step or continuously in time. Eigenvalues decide what happens.

Key idea

Discrete time $x_{t+1} = Ax_t$: stable (shrinks to zero) if every $|\lambda| < 1$; blows up if some $|\lambda| > 1$.

Continuous time $\dot{x} = Ax$: stable if every eigenvalue has *negative real part*; imaginary parts produce oscillations.

Why eigenvalues govern stability

Intuition

Eigenvalues are the growth rates of the natural “modes” of a system. Decompose any starting state into its eigenvector components; each then evolves independently, growing or decaying at its own rate. The largest eigenvalue eventually dominates and sets the long-run behaviour.

This is why the eigenvalues alone tell you whether a dynamical system settles, oscillates, or explodes — without solving anything in detail.

Example

A discrete system $x_{t+1} = Ax_t$ has eigenvalues $\lambda_1 = 0.5$ and $\lambda_2 = 1.2$. What happens over time?

Along the first eigenvector the state is multiplied by 0.5 each step, so that component decays to zero. Along the second it is multiplied by 1.2, growing by 20% each step. Because one $|\lambda| > 1$, the system is **unstable**: almost every starting state eventually grows without bound. It would be stable only if *both* eigenvalues had magnitude below 1.

Link to cognitive science

Recurrent neural networks and models of neural population dynamics are exactly systems of the form $\dot{x} = Ax$ (plus inputs and nonlinearities). Whether a memory persists, an activity pattern decays, or a network oscillates is governed by the eigenvalues of its connectivity matrix.

This also closes the loop with the differential-equations material elsewhere in the programme: the qualitative behaviour of a linear system is read straight off its eigenvalues.

Practice — try it

Stable or unstable?

1. A continuous system with eigenvalues -1 and -3 .
2. A continuous system with eigenvalues -2 and $+0.5$.
3. A discrete system with eigenvalues 0.9 and 0.4 .

Session 4 — what we built

- *Eigenvectors* are the directions a matrix only stretches, by factor λ ; symmetric matrices have real eigenvalues and orthogonal eigenvectors.
- *PCA* uses the eigenvectors of the covariance matrix as principal components, and the eigenvalues as the variances along them.
- The *SVD* is the general rotation–scale–rotation behind PCA.
- In *dynamics*, eigenvalues set the growth or decay of each mode, and thus stability.

Next: probability again — now armed with vectors and matrices.

Session 5 — Advanced Probability & Statistics

What today is about

This session fuses the probability of Sessions 1–2 with the linear algebra of Sessions 3–4 into the tools that models actually use. It is a recap and an elevation for students who already have some mathematics.

- *Bayesian inference* over a whole parameter.
- The *multivariate Gaussian* — where the covariance matrix meets PCA.
- *Maximum likelihood*: the one principle behind most model fitting.
- The *GLM*, *model comparison*, *information theory*, and a taste of *sampling*.

Bayes as a framework, not just a formula

In Session 1, Bayes updated the probability of a single hypothesis. As a *framework*, it updates a whole distribution over a continuous parameter.

Key idea

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{p(D)} \propto \underbrace{p(D | \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}.$$

We start with a prior distribution over θ , weight it by how well each θ predicts the data, and renormalise to get the posterior.

The answer is a distribution, not a point

Intuition

The Bayesian answer is never a single number — it is a whole distribution that carries its own uncertainty. A sharp posterior means the data pinned θ down; a broad one means we are still unsure.

Two common summaries: the **posterior mean** (the average belief), and the **MAP estimate** (the single most probable value, the peak of the posterior).

Beta–Binomial

We want the probability θ that a participant detects a faint stimulus. Put a uniform prior $\theta \sim \text{Beta}(1, 1)$ and observe 8 detections in 10 trials.

For a $\text{Beta}(a, b)$ prior and h successes in $h + t$ trials, the posterior is $\text{Beta}(a + h, b + t)$. Here it is $\text{Beta}(9, 3)$, with posterior mean $\frac{9}{12} = 0.75$ and MAP $\frac{8}{10} = 0.80$. The prior gently pulls the estimate away from the extreme, and the posterior's width honestly reports the remaining uncertainty.

Two facts that make Bayes practical

Useful fact

Sequential updating: today's posterior is tomorrow's prior — processing data all at once or one point at a time gives the *same* posterior, so beliefs can be updated online.

Credible interval: a range holding 95% of the posterior. Unlike a confidence interval, it *does* allow the statement “there is a 95% probability θ is in here”.

Link to cognitive science

Bayesian inference is the dominant formal language of contemporary cognitive modeling: perception as inference, cue combination, learning, and the “Bayesian brain” all cast cognition as computing posteriors. The prior encodes expectations, the likelihood encodes sensory evidence, and behaviour reflects the posterior.

The prior is a modeling choice

Common pitfall

The prior is a deliberate choice, not a nuisance to hide. With little data it matters; with abundant data the likelihood dominates and the posterior becomes insensitive to any reasonable prior.

Good practice is to state priors explicitly and check that conclusions are robust to sensible alternatives.

Practice — try it

Start from a uniform $\text{Beta}(1, 1)$ prior on a detection probability θ .

1. After 3 hits in 3 trials, what is the posterior?
2. What is its posterior mean?
3. You then see 1 hit in the next 2 trials — update again.

Several variables at once

Real data have many variables that must be modelled *jointly*.

Definition

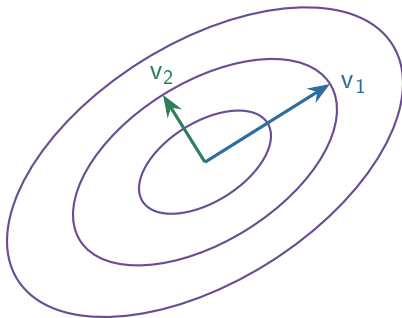
The **joint** distribution $p(x, y)$ describes variables together; the **marginal** $p(x) = \sum_y p(x, y)$ describes one alone; the **conditional** $p(x | y)$ describes one given another. They are linked by $p(x, y) = p(x | y) p(y)$.

The multivariate Gaussian

Key idea

$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ generalises the bell curve to many dimensions. It is defined by a **mean vector** $\boldsymbol{\mu}$ (the centre) and a **covariance matrix** $\boldsymbol{\Sigma}$ (the shape). Its contours of equal probability are ellipses whose axes are the *eigenvectors* of $\boldsymbol{\Sigma}$ and whose widths are set by its *eigenvalues* — exactly the PCA picture.

The covariance ellipse



Equal-probability contours of a 2-D Gaussian. The ellipse axes are the eigenvectors of the covariance matrix; their lengths grow with the eigenvalues. This is the same picture as PCA.

The object that needs both toolkits

Intuition

The covariance matrix is where statistics and linear algebra become the same subject. Statistics supplies its entries (variances and covariances); linear algebra reads its shape (eigenvectors and eigenvalues give the orientation and spread of the data ellipse).

This is the single most important unification in the course: “covariance”, “eigenvector”, and “principal component” turn out to be three views of one thing.

Why the Gaussian is so convenient

Useful fact

- **Closed** under marginals and conditionals: subsets and conditionals of a Gaussian are again Gaussian.
- For Gaussians, *uncorrelated* implies *independent* — so a diagonal covariance matrix means fully independent components.
- **Linear maps** keep it Gaussian: $Ax + b \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$, which is exactly what PCA's rotation uses.

The multivariate Gaussian in practice

Link to cognitive science

Multivariate Gaussians underlie models of neural noise correlations, Gaussian classifiers and linear discriminant analysis, Kalman filters for tracking and motor control, and the latent-variable models (factor analysis, probabilistic PCA) used to describe population activity.

Whenever someone says “noise is correlated across neurons”, a covariance matrix is doing the work.

Definition

The **maximum likelihood estimate** is the parameter value that makes the observed data most probable:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(D | \theta) = \arg \max_{\theta} \sum_i \log p(x_i | \theta),$$

where we maximise the *log*-likelihood (a sum, easier than a product).

Likelihood reverses the usual question

Intuition

A distribution asks: “given the parameters, how probable is each possible dataset?” Likelihood fixes the data we actually saw and asks the reverse: “which parameters would have made *this* most plausible?” Maximum likelihood simply picks that best-explaining parameter.

Taking the log turns the product over independent data points into a sum, which is what we actually optimise — and it does not move the maximum.

Maximum likelihood unifies what you know

Key idea

For Gaussian data, the MLE of the mean is just the *sample mean*. Fitting a regression line by *least squares* is exactly maximum likelihood under Gaussian noise. So “minimise squared error” and “maximise likelihood” are the same procedure in disguise — and in machine learning, the negative log-likelihood is the “loss” we minimise.

Deriving the obvious estimate the formal way

Example

A participant succeeds on k of n independent trials. Find the MLE of the success probability p .

The log-likelihood is $\ell(p) = k \ln p + (n - k) \ln(1 - p)$. Setting the derivative to zero:

$$\frac{k}{p} - \frac{n - k}{1 - p} = 0 \implies \hat{p} = \frac{k}{n}.$$

Maximum likelihood recovers the natural estimate — the observed success rate — and the same machinery works when the answer is *not* obvious.

Practice — try it

1. A neuron responds on 12 of 40 trials. What is the MLE of its response probability?
2. Two models: A says $p = 0.5$, B says $p = 0.7$. You see 7 heads in 10 flips. Which assigns higher likelihood?
3. Why maximise the *log*-likelihood rather than the likelihood itself?

Extending regression to other data types

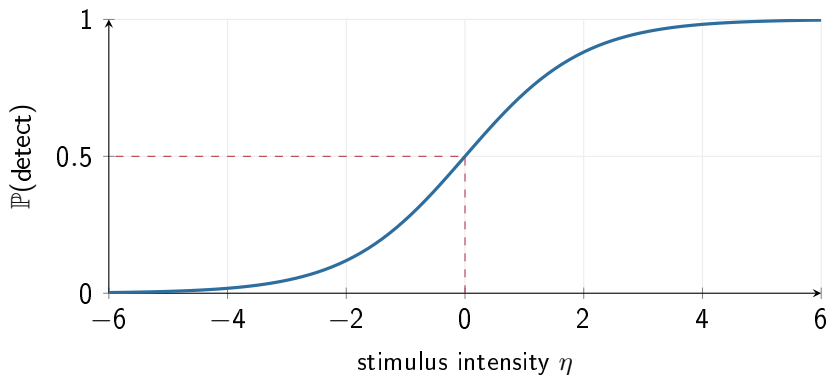
Regression assumed a straight line and Gaussian noise. Many outcomes are binary (correct/incorrect) or counts (spikes). The GLM extends regression to these while keeping its linear core.

Definition

A **Generalized Linear Model** keeps a linear predictor $\eta = \beta_0 + \beta_1 x_1 + \dots$, but passes it through a **link function**. For binary outcomes, *logistic regression* uses the logistic (sigmoid) link:

$$\mathbb{P}(y = 1 | x) = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}.$$

The psychometric function



The sigmoid squashes any real number into a probability in $[0, 1]$, producing the familiar S-shaped psychometric function.

What the link function does, and what coefficients mean

Intuition

The link is a translator. The linear part η can be any number, but a probability must live in $[0, 1]$. The sigmoid squashes the whole real line into that range, giving the smooth rise from 0 to 1 as stimulus intensity increases.

Coefficients are log-odds

$\ln \frac{\mathbb{P}(y = 1)}{\mathbb{P}(y = 0)} = \beta_0 + \beta_1 x_1 + \dots$, so each β_j is the change in log-odds per unit of x_j , and e^{β_j} is the **odds ratio**.

Link to cognitive science

Logistic regression *is* the psychometric function relating stimulus to detection probability. Poisson regression models spike counts as a function of stimuli (the linear–nonlinear–Poisson encoding model). The same framework yields decoding models that read stimuli off neural activity.

Master the GLM and a large family of analysis methods becomes one familiar object.

Practice — try it

A logistic model of “detected” has $\eta = \beta_0 + \beta_1 x$ with $\beta_0 = 0$.

1. At $\eta = 0$, what is the detection probability?
2. As x grows large, what does the probability approach?
3. If $\beta_1 = \ln 2$, by what factor do the odds of detection change per unit of x ?

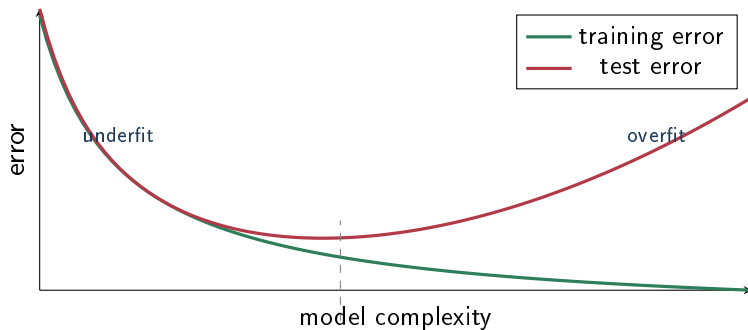
The most transferable skill in the course

A model that fits the training data perfectly may predict new data terribly. Guarding against this is what separates fitting from understanding.

The bias–variance trade-off

Too simple a model *underfits* (high bias — it misses real structure). Too complex a model *overfits* (high variance — it chases noise). Good modeling finds the middle: flexible enough to capture the signal, not so flexible that it memorises the noise.

Underfitting, overfitting, and the sweet spot



Training error always falls with complexity, but test error turns back up when the model starts fitting noise. The best complexity is where test error bottoms out.

Two practical tools

Key idea

Cross-validation: hold out part of the data, fit on the rest, and measure prediction error on the held-out part. This directly estimates generalisation — the gold standard.

Information criteria: score a model by fit *penalised* for complexity, $AIC = 2k - 2 \ln \hat{L}$ and $BIC = k \ln n - 2 \ln \hat{L}$. Lower is better.

Example

Two models fit to $n = 100$ points. Model 1: $k = 2$, $\ln \hat{L} = -50$. Model 2: $k = 4$, $\ln \hat{L} = -47$ (a little better, as a more flexible model must be).

$$\text{AIC}_1 = 104, \quad \text{AIC}_2 = 102 \Rightarrow \text{AIC picks Model 2,}$$

$$\text{BIC}_1 \approx 109.2, \quad \text{BIC}_2 \approx 112.4 \Rightarrow \text{BIC picks Model 1.}$$

BIC's heavier penalty ($\ln n \approx 4.6$ per parameter vs. AIC's 2) favours the simpler model. When criteria disagree, cross-validation is the tie-breaker.

The one mistake to never make

Common pitfall

Never judge a model only by how well it fits the data it was trained on — more parameters *always* fit training data better. The honest question is always out-of-sample prediction.

Practice — try it

1. Adding parameters lowers training error. Does that make the model better?
2. Two models with equal test error — which do you keep?
3. Which penalises complexity more for $n > 7$, AIC or BIC?

Definition

Entropy $H(X) = -\sum_x p(x) \log_2 p(x)$ is the average uncertainty (in bits). **KL divergence** $D_{\text{KL}}(p\|q) \geq 0$ measures how different q is from a reference p . **Mutual information** $I(X; Y)$ measures how much knowing Y reduces uncertainty about X .

Entropy — intuition and example

Intuition

Entropy is “how surprised do I expect to be?” A fair coin has entropy 1 bit (maximally uncertain); a two-headed coin has entropy 0 (no surprise). KL divergence is the extra cost of describing reality with the wrong distribution; mutual information is the information-theoretic cousin of correlation, but sensitive to *any* dependence.

Example

A coin with $p = 0.25$: $H = -0.25 \log_2 0.25 - 0.75 \log_2 0.75 \approx 0.81$ bits — less than the 1 bit of a fair coin, because the bias makes it more predictable.

A few facts, and why they matter for ML

Useful fact

The uniform distribution maximises entropy ($\log_2 m$ bits). Cross-entropy splits as $H(p, q) = H(p) + D_{\text{KL}}(p||q)$ — which is why minimising cross-entropy loss trains a classifier toward the true distribution. And $I(X; Y) = H(X) - H(X | Y) \geq 0$, equal to 0 exactly when X and Y are independent.

Link to cognitive science

Information theory quantifies how much a neuron tells you about a stimulus (mutual information), underlies the efficient-coding hypothesis, and supplies the loss functions of machine learning.

Practice — try it

1. What is the entropy of a fair six-sided die?
2. Which has more entropy: a fair coin, or a coin with $p = 0.9$?
3. If knowing Y makes X certain, what is $I(X; Y)$ in terms of $H(X)$?

How Bayesian models are actually computed

Definition

A **Markov chain** is a sequence of states in which the next state depends only on the current one (the *Markov property*). Its dynamics are captured by a **transition matrix** — so propagating the state forward is repeated matrix multiplication, a nice callback to linear algebra.

Sampling: replacing integrals with histograms

Intuition

Posteriors in realistic models are too complicated to write down or integrate. So instead of computing them, we *sample*. **Markov chain Monte Carlo** builds a chain that wanders through parameter space, spending time in each region in proportion to its posterior probability. The collected samples then approximate the posterior — histograms stand in for integrals.

Link to cognitive science

Sampling has even been proposed as a model of cognition itself: perhaps the brain represents uncertainty by drawing samples. Markov models also describe sequential behaviour and spike-train dynamics.

- **Bayes** updates a full distribution (prior $\rightarrow$ posterior; mean or MAP).
- The **multivariate Gaussian** unites both subjects through the covariance matrix, whose eigenstructure is PCA.
- **Maximum likelihood** is the fitting principle; least squares is a special case.
- The **GLM** extends regression; **model comparison** fights overfitting; **information theory** measures uncertainty; **MCMC** computes posteriors.

You now hold the mathematical vocabulary of modern cognitive modeling, computational neuroscience, machine learning, and data analysis.